Do LLMs Fail the Same Way? A Unified Study of Attention and Learning Failures in Large Language Models

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***Abstract*—Large Language Models (LLMs) perform strongly on many benchmark tasks, yet their failure behavior under structured stress remains poorly understood. In particular, it is unclear whether errors arising in attention-intensive settings and those arising in learning-intensive settings reflect separate weaknesses or a shared underlying vulnerability. This paper presents a unified empirical analysis of LLM failure patterns across two controlled benchmark families: an attention benchmark that varies noise type, target position, and conflict strength, and a hard learning benchmark that evaluates belief updates, delayed supervision, exception rules, and conflicting mappings. We evaluate four open-source models—Gemma, Llama, Qwen, and Mistral—and analyze performance using descriptive statistics, heatmaps, statistical testing, and classical machine learning. Three stable patterns emerge across both benchmark families: instructional interference is the most damaging condition, end-position placement sharply reduces accuracy, and model-level robustness remains broadly consistent across domains. A unified failure prediction model using only structural metadata reaches approximately 74.8% accuracy, while task type contributes comparatively little to prediction. These findings suggest that many attention and learning failures are shaped by a shared set of structural factors rather than by entirely separate task-specific mechanisms. The study contributes a unified analytical framing for understanding LLM reliability and offers evidence that many failures are systematic, predictable, and potentially diagnosable prior to deployment.**

***Index Terms*—large language models, failure analysis, robustness, attention, in-context learning, instruction interference, positional bias, predictive modeling**

I. Introduction

Large Language Models (LLMs) have demonstrated impressive capabilities across natural language understanding, generation, reasoning, and instruction following [1], [2]. Despite these advances, their behavior remains fragile in the presence of noisy prompts, conflicting instructions, delayed supervision, and complex contextual structure. While many evaluations emphasize average task performance, practical deployment requires understanding *how* models fail and whether those failures follow repeatable patterns.

A central open question is whether failures in different cognitive settings reflect distinct weaknesses or a shared structural limitation. Some tasks primarily stress *attention*: the model must identify relevant information while ignoring misleading context, irrelevant salience, or positional distractions. Other tasks primarily stress *learning*: the model must update rules, resolve contradictions, or apply delayed supervision across examples. These categories are often studied separately, which makes it difficult to determine whether their failure modes are fundamentally different or whether they arise from the same underlying vulnerabilities.

This paper addresses that gap by analyzing two benchmark families within a common framework. The first benchmark focuses on attention failures under controlled prompt perturbations such as random irrelevant noise, stale context, emotional salience, misleading context, and instructional interference. The second benchmark focuses on hard learning failures through conflicting mappings, belief updates, delayed supervision, and exception rules. Across both benchmarks, we examine how performance varies as a function of model identity, target position, conflict strength, and noise family.

Our central hypothesis is that many apparent LLM failures are not random or narrowly task-specific. Instead, they may be governed by a compact set of structural variables that recur across domains. To test this hypothesis, we combine descriptive analysis with predictive modeling. If a relatively simple model can predict whether an LLM will succeed or fail using only metadata such as noise type, target position, conflict strength, and model identity, then the observed failures are likely systematic rather than arbitrary.

The main contributions of this paper are as follows:

- We present a unified empirical analysis of attention and learning failures using a shared evaluation framework.
- We show that instructional interference and end-position placement consistently produce the largest degradations across both benchmark families.
- We demonstrate that robustness rankings across models remain broadly stable across domains.
- We show that a unified failure prediction model can explain a substantial fraction of the observed variance using only structural metadata.

Taken together, these findings suggest that failures across attention and learning tasks are often shaped by shared structural

factors rather than by unrelated benchmark-specific effects.

II. Related Work

LLM robustness has been studied through prompt sensitivity, adversarial instruction design, in-context learning instability, and long-context reasoning failures. Prior work has shown that model outputs can shift substantially when prompts include misleading content, instruction conflicts, distractors, or subtle formatting changes [3]. Studies of long-context behavior have also observed strong positional degradation effects when relevant information is placed away from the beginning of the prompt [4]. More broadly, LLMs have demonstrated strong few-shot and in-context capabilities, but these capabilities remain sensitive to prompt structure and reasoning setup [1], [2].

Much of the existing literature, however, isolates one failure category at a time. Studies of attention-related failure often focus on retrieval, distraction, salience, or context-window effects. Studies of learning-related failure emphasize adaptation, belief revision, or rule induction. This separation is analytically useful, but it leaves open a deeper question: do these failure categories ultimately depend on the same structural variables?

This work complements existing robustness research by treating attention and learning failures as comparable outcomes within one framework. Rather than proposing a new model architecture, the goal is diagnostic: to identify whether the same variables recur as dominant predictors across task families.

III. Problem Formulation

We formalize each model response as a binary outcome. Let

$$Y_i \in \{0, 1\} \tag{1}$$

indicate whether the model answered item i correctly, where $Y_i = 1$ denotes success and $Y_i = 0$ denotes failure.

Each item is associated with a structured feature vector

$$X_i = \big(t_i, n_i, p_i, c_i, m_i\big), \tag{2}$$

where:

- t_i is the task type (attention or learning),
- n_i is the noise family,
- p_i is the target position,
- c_i is the conflict strength,
- m_i is the model identity.

The core question of this work is whether

$$P(Y_i = 1 \mid X_i) \tag{3}$$

can be explained by a shared set of structural features across both benchmark families.

We also define model-level accuracy for a subset S of benchmark items as

$$\mathrm{Acc}(S) = \frac{1}{|S|} \sum_{i \in S} Y_i. \tag{4}$$

To analyze overlap in failure behavior, we consider whether the same features that reduce accuracy in one domain also reduce accuracy in the other. In the predictive setting, this becomes a supervised classification problem in which the target is Y_i and the input is X_i. If a relatively simple classifier can estimate Y_i with strong accuracy, then failure behavior is structurally organized rather than random.

IV. Benchmark Design and Data

The study uses two benchmark families generated from controlled prompt conditions.

A. Attention Benchmark

The attention benchmark evaluates how well models preserve the relevant signal when prompts contain structured distraction. The controlled variables include:

- **noise family:** no noise, random irrelevant noise, stale context, emotional salience, misleading context, and instructional interference,
- **target position:** none, beginning, middle, or end,
- **conflict strength:** the degree to which distracting or competing information directly contradicts the target signal.

The purpose of this benchmark is to isolate failures in selective attention, instruction prioritization, and resistance to misleading prompt structure.

B. Hard Learning Benchmark

The hard learning benchmark evaluates whether models can learn, revise, or maintain rules under difficult contextual conditions. The major task families are:

- belief update,
- delayed supervision,
- exception rule,
- conflicting mapping.

This benchmark also includes structured noise, positional manipulation, and conflict variation so that it can be analyzed using the same framework as the attention benchmark.

C. Dataset Composition

The combined analysis includes 2,720 model-item evaluations:

- 2,240 evaluations from the attention benchmark,
- 480 evaluations from the hard learning benchmark.

At the model level, the attention benchmark contains 560 evaluated items per model, while the hard learning benchmark contains 120 evaluated items per model. This balanced per-model setup supports consistent cross-model comparison within each benchmark family.

D. Models Evaluated

We evaluate four open-source LLM families:

- Gemma,
- Llama,
- Qwen,
- Mistral.

These models were selected to provide variation in observed robustness while remaining comparable under a common evaluation pipeline. Each model's outputs are aggregated into unified analysis tables for within-benchmark and cross-benchmark comparison.

V. Methodology

The analysis proceeds in four stages.

A. Descriptive Analysis

We first compute aggregate accuracy grouped by model, noise family, target position, and learning task family. This establishes the primary empirical trends.

B. Global Visualization

We then construct heatmaps to compare failure patterns compactly across models and task settings. These views are useful for identifying stable cross-model trends that may not be obvious from individual bar plots.

C. Predictive Failure Modeling

To test whether failures are systematic, we train classical machine learning models using only structured metadata. The features are task type, noise family, target position, conflict strength, and model identity. We evaluate:

- Random Forest,
- Logistic Regression.

Prediction is performed first separately for each benchmark family and then on the combined dataset.

D. Statistical Testing

Finally, we run significance tests to assess whether observed differences are statistically reliable. These include:

- ANOVA across models,
- ANOVA across noise families,
- ANOVA across target positions,
- comparison between attention and learning outcomes.

VI. Results: Attention Benchmark

A. Accuracy by Model

The attention benchmark reveals a strong separation in robustness across models. Gemma achieves the highest aggregate accuracy at 0.7839, followed by Qwen at 0.7446 and Llama at 0.7179, while Mistral trails substantially at 0.5268. The gap between Gemma and Mistral exceeds 25 percentage points, indicating that robustness under attention stress differs sharply across model families.

B. Accuracy by Noise Family

Most noise categories result in relatively high and clustered accuracy, including misleading context (0.8317), no noise (0.8171), emotional salience (0.7955), and stale context (0.7795). Random irrelevant noise causes only moderate degradation (0.7367). By contrast, instructional interference causes a dramatic collapse in performance, making it the most destructive attention-side condition.

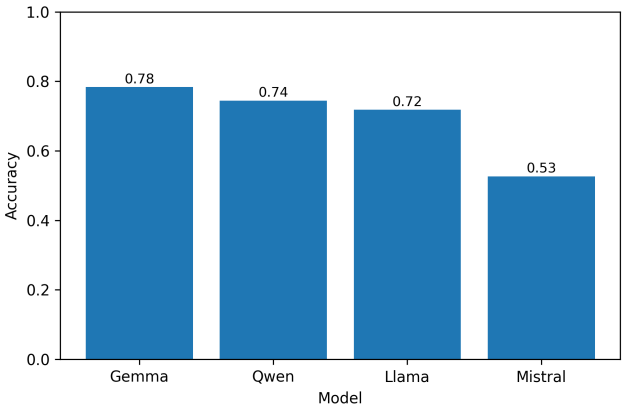

Fig. 1. Model accuracy under the attention benchmark. Gemma achieves the highest performance, while Mistral exhibits substantial degradation under structured noise conditions.

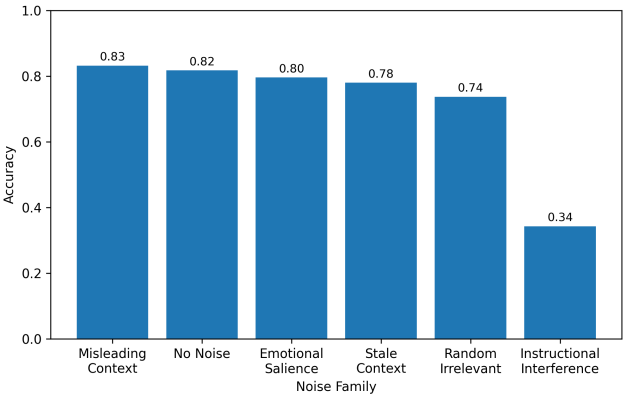

Fig. 2. Accuracy under different attention-side noise conditions. Instructional interference causes the largest performance drop, while the other noise families remain comparatively less harmful.

C. Accuracy by Target Position

Target placement exerts a major influence on performance. Accuracy remains high when the relevant signal appears at the beginning (0.7298) or middle (0.7785) and is highest when no positional manipulation is used (0.8171). However, when the signal is pushed to the end of the prompt, accuracy drops sharply to 0.2040. This large decline indicates a strong positional bias under attention stress.

VII. Results: Hard Learning Benchmark

A. Accuracy by Model

The hard learning benchmark preserves a similar model ordering. Gemma again performs best at 0.9250, followed by Llama at 0.8333 and Qwen at 0.7833, while Mistral falls well behind at 0.4583. The gap between the strongest and weakest model now exceeds 45 percentage points.

B. Accuracy by Task Family

Performance varies substantially by learning task family. Models do best on belief update (0.8917) and delayed supervision (0.8250), decline on exception rule tasks (0.7167), and

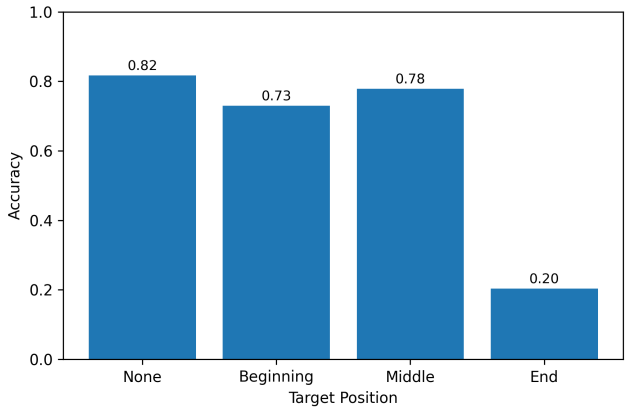

Fig. 3. Accuracy by target position in the attention benchmark. Performance remains high when relevant information appears early or mid-prompt, but drops sharply when placed at the end, revealing a strong positional bias.

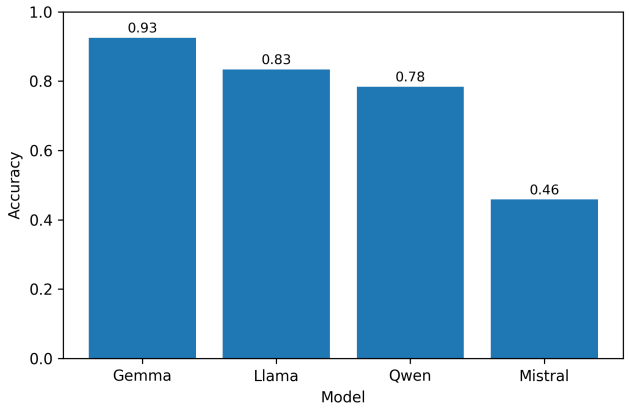

Fig. 4. Model accuracy under the hard learning benchmark. Gemma achieves the highest performance, while Mistral shows substantial degradation, consistent with attention benchmark trends.

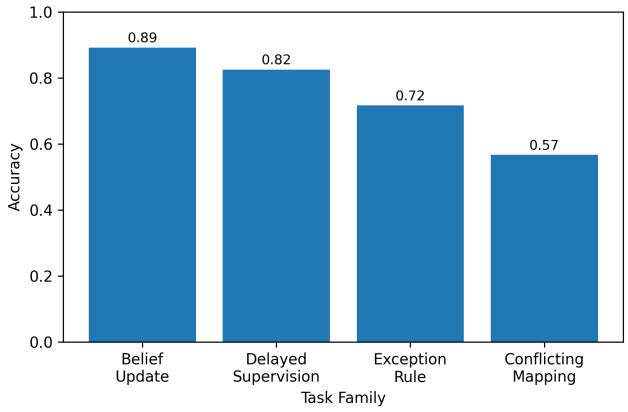

Fig. 5. Accuracy across hard learning task families. Performance is highest for belief updates and lowest for conflicting mappings, indicating that models struggle most with resolving conflicting rules.

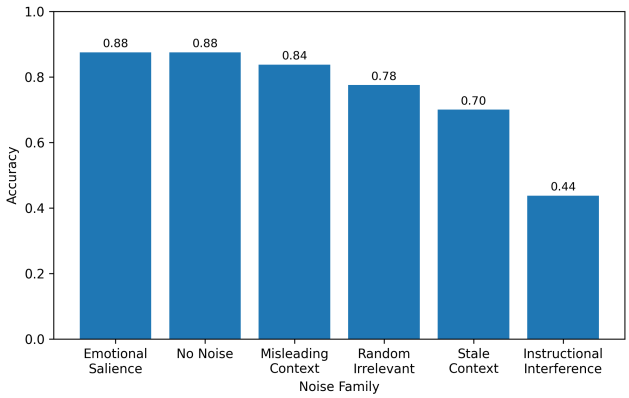

Fig. 6. Accuracy under noise conditions in the hard learning benchmark. Instructional interference again produces the largest degradation, consistent with attention benchmark results.

struggle most on conflicting mapping (0.5667). This suggests that learning failures are concentrated in contexts that require conflict resolution rather than in all revision tasks equally.

C. Noise and Position Effects

The learning benchmark mirrors the attention benchmark in two critical ways. First, instructional interference again acts as the dominant negative condition, with accuracy falling to 0.4375 compared with much higher values for emotional salience (0.8750), no noise (0.8750), and misleading context (0.8375). Second, target position again matters strongly: when the critical signal appears at the end, accuracy drops to 0.4375, compared with 0.8750 when no positional manipulation is applied and 0.7969 when the signal appears near the beginning.

VIII. Unified Analysis Across Both Domains

When the two benchmark families are merged into a unified analysis table, two important results emerge.

First, for the stronger models, aggregate performance on hard learning tasks exceeds aggregate performance on attention tasks. Gemma improves from 0.7839 in attention to 0.9250 in learning, while Llama improves from 0.7179 to 0.8333. This suggests that attention-heavy settings may in some cases be harder than conceptually richer learning tasks.

Second, and more importantly, the *relative ordering of models remains stable* across domains. Gemma remains the strongest, Mistral remains the weakest, and the intermediate models preserve broadly similar positions. This indicates that robustness is not narrowly benchmark-specific.

The heatmaps reinforce three broad conclusions: instructional interference is the strongest source of degradation, end-position information consistently hurts performance, and model-level differences remain stable across domains.

IX. Failure Prediction with Classical Machine Learning

A major goal of this study is to determine whether LLM failures are predictable using only structural metadata. To test this, we train classical machine learning models on benchmark descriptors rather than prompt semantics.

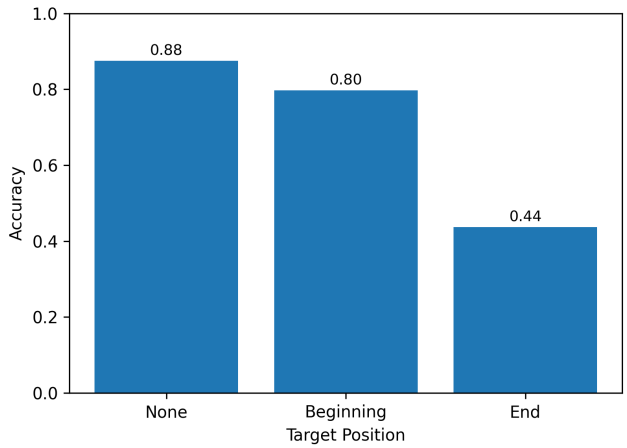

Fig. 7. Accuracy by target position in the hard learning benchmark. Performance declines when the relevant signal is delayed, with the strongest drop occurring when it appears at the end.

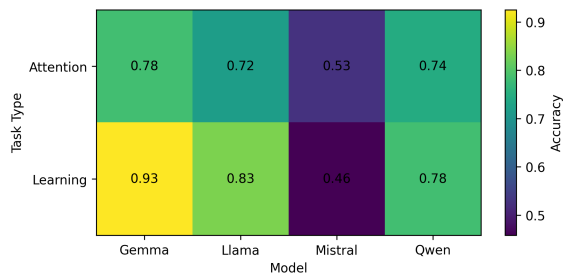

Fig. 8. Heatmap comparing model performance across attention and learning benchmarks. Model rankings remain broadly stable across domains, with Gemma consistently strongest and Mistral consistently weakest.

A. Benchmark-Specific Prediction

For the attention benchmark, a Random Forest model reaches 0.7478 accuracy using noise type, target position, conflict strength, and model identity. The model predicts correct cases more easily than failure cases, but the overall result is notable because no prompt text is used.

For the hard learning benchmark, Random Forest reaches 0.8229 accuracy. This higher performance suggests that learning benchmark outcomes are even more structurally predictable than attention benchmark outcomes.

B. Unified Failure Modeling

We then train a combined model over both benchmark families. The unified Random Forest reaches 0.7482 accuracy. Logistic Regression achieves the same overall accuracy, which is especially informative: because Logistic Regression is linear and much simpler, the core failure patterns appear to be not only systematic but also relatively simple in structure.

The strongest unified predictors are:

- instructional interference (feature importance: 0.2569),
- model identity, especially Mistral (0.1671),
- end-position prompts (0.1632),
- strong conflict strength (0.0567).

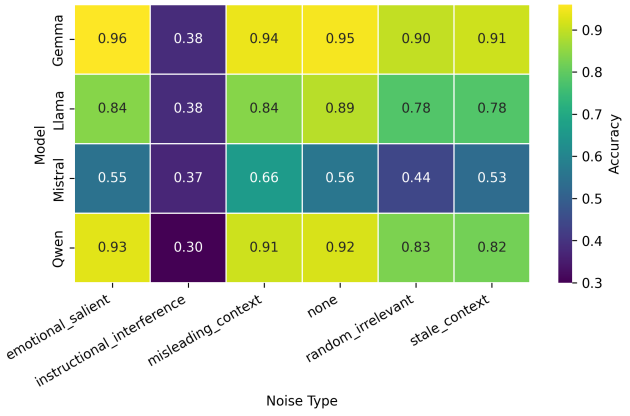

Fig. 9. Heatmap of model performance across noise families. Instructional interference consistently produces the lowest accuracy across all models, highlighting it as the dominant source of failure.

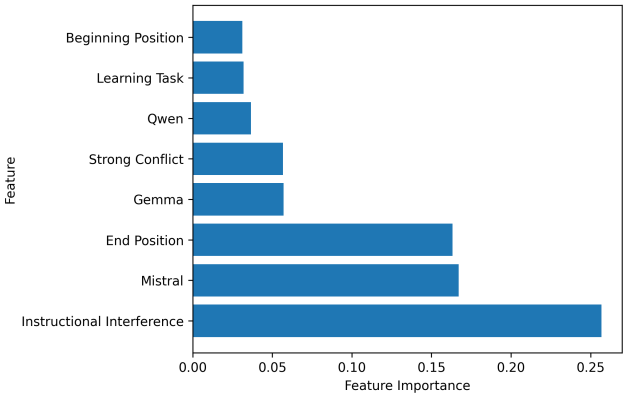

Fig. 10. Feature importance from the unified failure prediction model. Instructional interference, model identity, and end-position placement emerge as the strongest predictors, while task type contributes minimally.

By comparison, task type contributes only modestly (roughly 0.03), suggesting that whether a task belongs to the attention or learning benchmark is far less important than its structural composition.

X. Statistical Validation

The descriptive and predictive patterns are supported by statistical testing.

ANOVA across models yields a highly significant result ($F = 57.99$, $p < 10^{-35}$), confirming that differences in model robustness are not attributable to random variation. The comparison between attention and learning outcomes is also statistically significant ($t = -2.47$, $p = 0.0136$), although the effect size is relatively small (Cohen's $d = -0.127$). This reinforces the idea that the distinction between attention and learning is less important than the structural variables embedded within each task. ANOVA across target positions is also highly significant ($F = 139.51$, $p < 10^{-83}$), and ANOVA across noise families confirms a strong effect of noise type ($F = 100.39$, $p < 10^{-96}$)).

TABLE I
KEY STATISTICAL TEST RESULTS

Test	Statistic	Interpretation
ANOVA across models	$F = 57.99$	Strong differences across models, indicating that model architecture significantly affects robustness under structured stress.
Attention vs. learning t-test	$t = -2.47,\ p = 0.0136$	Statistically significant difference between attention and learning outcomes, though the magnitude of the difference is modest.
Attention vs. learning effect size	$d = -0.127$	Small practical difference between attention and learning benchmarks.
ANOVA across target positions	$F = 139.51$	Strong positional effect, indicating that the placement of critical information substantially influences correctness.
ANOVA across noise families	$F = 100.39$	Strong effect of noise type on model performance, with instructional interference causing the most significant degradation.

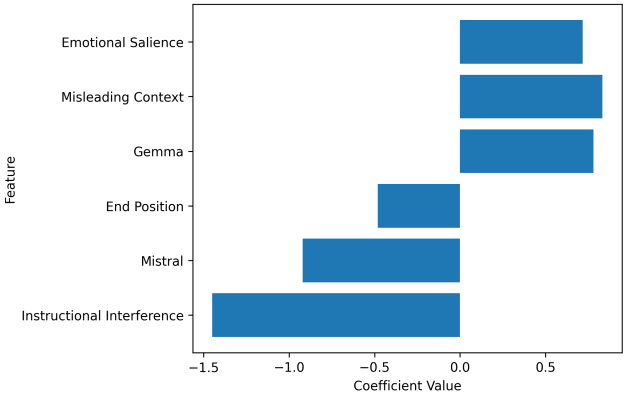

Fig. 11. Logistic regression coefficients indicating directional effects on prediction. Instructional interference, Mistral, and end-position placement negatively impact performance, while Gemma and certain noise conditions have positive effects.

XI. DISCUSSION

The results point to a broader interpretation of LLM failure. Rather than behaving as unrelated benchmark-specific errors, many failures appear to emerge from a shared structural vulnerability profile.

The most important evidence for this claim is the consistency of the dominant predictors. Instructional interference harms performance across both attention and learning tasks. End-position placement creates a substantial drop in both settings. Model identity remains a stable determinant of robustness regardless of benchmark type. Finally, predictive models recover these same factors using only high-level metadata.

These observations suggest that some LLM weaknesses may be better understood as failures of structured control under competing contextual demands. When instructions conflict, when the signal is delayed, or when contextual salience competes with relevance, models become less reliable in systematic ways. This is important because such conditions are common in real-world deployments, including retrieval-augmented generation, long-context assistants, and multi-step interaction systems.

The study also carries a practical implication: structural diagnostics may help identify risky prompt configurations before deployment. If failure can be predicted from metadata, then system designers may be able to screen prompts, rerank contexts, or redesign interfaces to reduce avoidable errors.

A. Limitations

This study has several limitations. First, it evaluates only four open-source model families. Second, the analysis focuses on benchmark-generated conditions rather than naturally occurring user prompts. Third, the work measures final correctness rather than richer behaviors such as calibration, uncertainty expression, or partial reasoning quality. Fourth, the reference base is intentionally focused and not exhaustive; a broader journal-length treatment would benefit from deeper engagement with adjacent literature on robustness, long-context reasoning, and in-context learning dynamics. Future work should test whether the same structural predictors transfer to larger proprietary models, multimodal systems, and longer real-world workflows.

XII. CONCLUSION

This paper presented a unified analysis of LLM failures across attention and learning benchmarks. The results show that failures are neither random nor strongly task-specific. Instead, they are systematically governed by a compact set of structural variables, especially instructional interference, positional bias, conflict strength, and model architecture.

Across both benchmark families, these variables consistently determine whether models succeed or fail. The unified predictive models further show that many failure outcomes are not only systematic but also learnable from simple metadata alone. This supports the broader conclusion that attention and learning failures are often shaped by a shared set of structural factors rather than by completely separate failure processes.

Understanding these mechanisms is essential for improving LLM reliability in practical systems. By moving from average accuracy toward structured failure analysis, we gain a more actionable view of where current models are robust, where they break, and how future systems may be made more dependable.

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